

A SCALE-TENSOR PROOF OF AN ABP ESTIMATE ON STRATIFIED CARNOT GROUPS

ABSTRACT. We develop a large- p ABP estimate for arbitrary bounded measurable symmetric horizontal coefficient matrices on a stratified Carnot group. The regularizer is an all-layer, time-dependent left-invariant diffusion. Its exact tensor lift is renormalized by its finite-dimensional zero-order moment evolution. The normalized nonhorizontal blocks have zero scale-boundary data and positive scale damping. Their energies at layer k have constants depending only on smoothing weights strictly below k . Two integrations by parts pair the remaining fluxes with scalar gradient dissipation and yield an inverse current-layer smoothing factor. A finite graded parameter selection, followed by a small choice of $q - 1$, closes the forward adjoint entropy inequality. The elliptic maximum estimate follows by resolvent duality, a horizontal quadratic barrier, and smooth approximation of the coefficients.

1. THE RESULT AND THE ROLE OF THE NEW CORRECTION

Let $\mathfrak{g} = V_1 \oplus \cdots \oplus V_s$ be a stratified Lie algebra, $r = \dim V_1$, and $D = \sum_{j=1}^s j \dim V_j$. Fix orthonormal bases $Z_{j,\alpha}$ of the layers and write $X_\alpha = Z_{1,\alpha}$. All fields are left invariant. Haar measure is understood. Let

$$L_A u = \sum_{\alpha,\beta=1}^r a_{\alpha\beta} X_\alpha X_\beta u.$$

Symmetry of A means that this is contraction with the symmetrized horizontal Hessian. A fixed left-invariant homogeneous distance defines the balls below.

Theorem 1. *For every stratified Carnot group $\mathbb{G}$ and $E \geq 1$, there is a finite $p_0 = p_0(\mathbb{G}, E) > 1$ with the following property. Let $\Omega \subset B_R(g_0)$ be bounded and open. Suppose $A : \Omega \rightarrow \text{Sym}_r$ is Borel measurable, $\lambda I \leq A \leq \Lambda I$ almost everywhere, and $E = \Lambda/\lambda$. If $u \in C^2(\Omega) \cap C(\overline{\Omega})$ and $L_A u \geq f$ almost everywhere, then, for $p_0 \leq p \leq \infty$,*

$$(1.1) \quad \sup_{\Omega} u \leq b + \frac{C(\mathbb{G}, E, p)}{\lambda} R^{2-D/p} \|f^{-1} \mathbf{1}_{\{u>b\}}\|_{L^p(\Omega)}, \quad b = \sup_{\partial\Omega} u^+.$$

The constants do not involve coefficient derivatives or a modulus of continuity. No lattice or boundary regularity assumption is required.

The common-flow entropy calculation comes from the supplied kinetic and Heisenberg notes [1, 2]. The additional mechanism here is to remove the full tensor's zero-order moment evolution before estimating its higher blocks. This is different from either discarding the tensor commutator or requiring the complete regularized tensor to be pointwise elliptic.

Normalize $\lambda = R = 1$ until the final section. Initially A is smooth, $I \leq A \leq EI$, and is constant outside a compact set. Let

$$(1.2) \quad m_t = L_A^* m = \sum_{\alpha,\beta} X_\alpha X_\beta (a_{\alpha\beta} m), \quad m > 0, \quad \int m = 1.$$

The a priori estimates are uniform over these smooth coefficients. Smooth positive data with sufficiently rapid polynomial decay can be used initially; Section 8 explains the analytic reduction and approximation.

2. THE SCALAR REGULARIZER AND THE EXACT ENTROPY RULE

Choose $a_1 = 1$ and $a_j \geq 1$, $2 \leq j \leq s$, to be fixed before choosing q . Put

$$(2.1) \quad \mathcal{A}_h = \frac{1}{2} \sum_{j=1}^s a_j h^{j-1} \sum_{\alpha} Z_{j,\alpha}^2, \quad \partial_h S_h = \mathcal{A}_h S_h, \quad S_0 = I.$$

The evolution is nonautonomous; S_h is not being identified with $\exp(h\mathcal{A}_h)$. It is positive and preserves mass. It is a convolution regularizer with a smooth positive kernel K_h , with convolution order fixed by (2.1). Homogeneous scaling gives

$$(2.2) \quad K_h(g) = h^{-D/2} K_1(\delta_{h^{-1/2}} g), \quad \int K_h = 1, \quad \|S_h\|_{1 \rightarrow \infty} \leq C_{\mathbb{G}} h^{-D/2}.$$

The last constant can be taken independent of the $a_j \geq 1$. Indeed, the horizontal Nash inequality and

$$\frac{d}{dh} \|S(h, h_0)v\|_2^2 = - \sum_j a_j h^{j-1} \|\nabla_j S(h, h_0)v\|_2^2 \leq -\|\nabla_H S(h, h_0)v\|_2^2$$

give the $1 \rightarrow 2$ bound. Applying the same argument to the reverse-time adjoint evolution gives $2 \rightarrow \infty$; composition gives (2.2). The horizontal Nash inequality follows from the fixed-sub-Laplacian heat bound by $\|v\|_2^2 \leq h\|\nabla_H v\|_2^2 + Ch^{-D/2}\|v\|_1^2$ and optimization. Only fixed-operator heat facts are used here; see [3].

At a fixed physical time, put

$$\rho = S_h m, \quad g = \log \rho, \quad q = 1 + \varepsilon, \quad 0 < \varepsilon \leq \frac{1}{2}, \quad Q(t, h) = \int \rho^q.$$

For a scalar or vector v , define

$$(2.3) \quad \mathcal{D}_q(v) = \sum_{j,\alpha} a_j h^{j-1} \int \rho^q \left[\frac{2-q}{q} |Z_{j,\alpha} v|^2 \right.$$

$$(2.4) \quad \left. + \frac{q\varepsilon}{2} \left| v Z_{j,\alpha} g + \frac{2}{q} Z_{j,\alpha} v \right|^2 \right].$$

If $U_h = \mathcal{A}_h U + F$ and $v = U/\rho$, then

$$(2.5) \quad \frac{d}{dh} \int \rho^{q-2} |U|^2 = -\mathcal{D}_q(v) + 2 \int \rho^{q-2} \langle U, F \rangle.$$

In addition to $\mathcal{D}_q(v) \geq \frac{1}{3} \sum a_j h^{j-1} \int \rho^q |Z_j v|^2$, we will repeatedly use the exact second form

$$(2.6) \quad \mathcal{D}_q(v) = \sum_{j,\alpha} a_j h^{j-1} \int \rho^q \left[|Z_{j,\alpha} v + \varepsilon v Z_{j,\alpha} g|^2 + \frac{\varepsilon(1-\varepsilon)}{2} |v|^2 (Z_{j,\alpha} g)^2 \right].$$

This is why no loss $1/(q-1)$ is needed in the source estimates below.

Write

$$(2.7) \quad I_j(h) = \sum_{\alpha} \int \rho^q (Z_{j,\alpha} g)^2, \quad J_j = \int_0^1 h^{j-1} I_j(h) dh,$$

$$(2.8) \quad L_j = \sum_{\alpha} \int_0^1 h^j \mathcal{D}_q(Z_{j,\alpha} g) dh, \quad J_* = \sum_j J_j, \quad L = \sum_j L_j.$$

Then

$$(2.9) \quad Q(t, 0) - Q(t, 1) = \frac{q\varepsilon}{2} \sum_j a_j J_j.$$

3. UNIFORM SCALAR ESTIMATES AND FINITE PARAMETER SELECTION

All constants in this section are uniform for $1 < q \leq 3/2$.

Lemma 2. *Given finitely many polynomials $P_k(a_1, \dots, a_{k-1})$ with nonnegative coefficients, and $\delta > 0$, the weights can be chosen so that*

$$(3.1) \quad J_* \leq 2J_1, \quad L \leq C_{\mathbb{G}} J_1, \quad \sum_{k=2}^s P_k(a_{<k}) J_k \leq \delta J_1.$$

One may impose simultaneously any finite collection of conditions $P_k(a_{<k})/a_k \leq \delta$. The constant $C_{\mathbb{G}}$ does not depend on the values of the weights selected in this way or on q .

Proof. For $U \in V_i$, $V \in V_j$, write $g_U = Ug$. In the entropy of $U\rho$, the commutator with $a_j h^{j-1} V^2/2$ has source

$$(3.2) \quad -2a_j h^{j-1} \int \rho^q (Vg_U + \varepsilon g_V g_U) [U, V] g + a_j h^{j-1} \int \rho^q g_U [[U, V], V] g.$$

This follows from $[U, V^2] = 2V[U, V] + [[U, V], V]$ and one integration by parts. Use (2.6), multiply by h^i , integrate, and discard the nonnegative upper endpoint. Summing the basis fields gives

$$(3.3) \quad L_i \leq C_{\mathbb{G}} \left[iJ_i + \sum_{j:i+j \leq s} a_j J_{i+j} + \sum_{j:i+2j \leq s} a_j \sqrt{J_i J_{i+2j}} \right].$$

Represent a basis of V_k by fixed linear combinations of $[X, Y]$, $X \in V_1$, $Y \in V_{k-1}$. The identity

$$\int \rho^q (Z_k g)^2 = \sum c \int \rho^q [(Xg)Y(Z_k g) - (Yg)X(Z_k g)]$$

has no derivative-of-weight remainder. Integrating with weight h^{k-1} and commuting Y and X to the right of Z_k yields

$$(3.4) \quad J_k \leq C_{\mathbb{G}} \left[\sqrt{J_1 (L_{k-1}/a_k + J_{2k-1})} + \sqrt{J_{k-1} (L_1/a_k + J_{k+1})} \right],$$

where $J_l = 0$ for $l > s$. The weights match exactly: the Z_k -dissipation in L_{k-1} has scale power h^{2k-2} , and that in L_1 has power h^k .

Here is the parameter argument. If $J_* > 0$, set $z_k = J_k/J_*$ and $\ell = L/J_*$. Using $z_k \leq 1$ in (3.4) and descending from $k = s$ gives a finite sum of bounds

$$(3.5) \quad z_k \leq C_{\mathbb{G}} \sum_{r=k}^s \sum_{\theta \in \Theta_{kr}} (\ell/a_r)^\theta, \quad 2^{-s} \leq \theta \leq \frac{1}{2}.$$

The finite sets Θ_{kr} depend only on the step. Equation (3.3), summed in i , gives

$$\ell \leq C_{\mathbb{G}} \left[1 + \sum_{j < k} a_j (z_k + \sqrt{z_k}) \right].$$

Take $a_1 = 1$ and, for $k \geq 2$,

$$(3.6) \quad a_k = M^{B^{k-2}}.$$

Choose the integer B sufficiently large compared with 2^s and the degrees of all the prescribed polynomials. Inserting (3.5) gives

$$\ell \leq C_{\mathbb{G}} \left[1 + M^{-c_s} \sum_{\theta \in \Theta} \ell^\theta \right], \quad 0 < \theta < 1, \quad c_s > 0.$$

For sufficiently large M , this gives $\ell \leq C_{\mathbb{G}}$, independently of M . Then (3.5) gives $z_k \leq C_{\mathbb{G}} a_k^{-c_s}$, after reducing c_s if needed. Since $a_k = a_{k-1}^B$, increasing B first and then M makes all $P_k(a_{<k})z_k$ and $P_k(a_{<k})/a_k$ as small as prescribed. Also $\sum_{k \geq 2} z_k \leq 1/2$, so $J_* \leq 2J_1$ and $L \leq C_{\mathbb{G}} J_1$. The case $J_* = 0$ is immediate. For $s = 1$, there are no higher layers and $L_1 \leq J_1$ directly. $\square$

This lemma is an existence statement for a separated hierarchy, not a claim that raw gradient energies have uniform bounds for every arbitrary unordered choice of diffusion weights. Its uniform constant is what permits subsequent weights to be made larger without restarting the estimates.

4. THE EXACT TENSOR LIFT AND REMOVAL OF ZERO-ORDER MOMENTS

Use one index α for the full homogeneous basis and write d_α for its degree. For a symmetric tensor M on $\mathfrak{g}$, set

$$\mathcal{H}M = \sum_{\alpha, \beta} Z_\alpha Z_\beta M_{\alpha\beta}, \quad C_\alpha = \text{ad}_{Z_\alpha}, \quad \mathcal{C}_\alpha M = C_\alpha M + M C_\alpha^T.$$

A direct expansion gives

$$(4.1) \quad [Z_\alpha, \mathcal{H}] = \mathcal{H}C_\alpha.$$

Therefore the exact tensor scale generator is

$$(4.2) \quad \begin{aligned} \mathcal{D}_h^{\text{ten}} &= \frac{1}{2} \sum_{\alpha} a_{d_\alpha} h^{d_\alpha-1} (Z_\alpha + \mathcal{C}_\alpha)^2 \\ &= \mathcal{A}_h + \sum_{\alpha} a_{d_\alpha} h^{d_\alpha-1} \mathcal{C}_\alpha Z_\alpha + \mathcal{V}_h, \end{aligned}$$

$$(4.3) \quad \mathcal{V}_h = \frac{1}{2} \sum_{\alpha} a_{d_\alpha} h^{d_\alpha-1} \mathcal{C}_\alpha^2.$$

Let A_H be the horizontal embedding of A , and solve

$$N_h = \mathcal{D}_h^{\text{ten}} N, \quad N(0) = m A_H.$$

Equation (4.1) implies

$$(4.4) \quad \mathcal{A}_h \mathcal{H} = \mathcal{H} \mathcal{D}_h^{\text{ten}}, \quad S_h \mathcal{H}(m A_H) = \mathcal{H} N, \quad \rho_t = \mathcal{H} N.$$

There is no residual commutator in the physical equation.

Define a finite-dimensional moment evolution

$$(4.5) \quad \mathbf{M}'_h = \mathcal{V}_h \mathbf{M}_h, \quad \mathbf{M}_0 = I, \quad \widehat{N} = \mathbf{M}_h^{-1} N.$$

Each C_α raises degree, so $\mathbf{M}_h$ and $\mathbf{M}_h^{-1}$ are finite polynomials in h and the weights. This removes the zero-order forcing:

$$(4.6) \quad \widehat{N}_h = \mathcal{A}_h \widehat{N} + \sum_{\alpha} a_{d_\alpha} h^{d_\alpha-1} \mathbf{M}_h^{-1} \mathcal{C}_\alpha \mathbf{M}_h Z_\alpha \widehat{N}.$$

The moment map preserves the positive-semidefinite cone. Indeed,

$$e^{\tau \mathcal{C}_\alpha^2/2} M = \mathbb{E}[e^{\sqrt{\tau} \xi C_\alpha} M e^{\sqrt{\tau} \xi C_\alpha^T}], \quad \xi \sim N(0, 1),$$

and products and time limits preserve this property. Put

$$D_h^{\text{gr}} = \text{diag}(h^{(d_\alpha-1)/2}).$$

For horizontal input M_H ,

$$(4.7) \quad \mathbf{M}_h(M_H) = D_h^{\text{gr}} \mathbf{M}_1(M_H) D_h^{\text{gr}}.$$

The $V_k \times V_k$ block of $\mathbf{M}_1(I_H)$ is bounded by a polynomial in $a_1, \dots, a_{k-1}$. Its horizontal block is exactly I_r .

5. THE LAYERWISE TENSOR ENERGY WITH STRICT WEIGHT INDEPENDENCE

For an unordered tensor block $\alpha = (i, k)$, $i \leq k$, put

$$n_\alpha = i + k - 2, \quad B_\alpha = h^{-n_\alpha/2} \widehat{N}_{ik} / \rho.$$

The horizontal block is

$$B_0 = B_{11} = S_h(Am) / \rho, \quad I \leq B_0 \leq EI.$$

It has the exact common-flow estimate

$$(5.1) \quad F_0(h) = \int \rho^q |B_0|^2, \quad F'_0 = -\mathcal{D}_q(B_0), \quad \int_0^1 \mathcal{D}_q(B_0) dh \leq rE^2 Q(t, 0).$$

Order the remaining blocks by increasing total degree, refining ties in any way. In (4.6), a source into B_α has the form

$$(5.2) \quad c_{\alpha\beta j}(a_{<k}) h^{j/2-1} (Z_j B_\beta + B_\beta Z_j g), \quad n_\beta < n_\alpha, \quad \max \beta \leq k, \quad j < k.$$

The coefficients are fixed linear maps and polynomials in weights strictly below k . To verify the power of h , a conjugated term with degree increment $n_\alpha - n_\beta$ has, before normalization, coefficient $h^{(n_\alpha - n_\beta + j)/2-1}$. Normalization leaves $h^{j/2-1}$. The restriction $j < k$ follows because ad_{Z_j} raises a nonzero input index by j , so its output index is at least $j + 1$. The same argument applies to every factor in the moment maps.

In addition to the scalar quotient generator, B_α has the drift $-n_\alpha B_\alpha / (2h)$. Set $F_\alpha = \int \rho^q |B_\alpha|^2$. Its exact energy identity is

$$(5.3) \quad F'_\alpha + \mathcal{D}_q(B_\alpha) + \frac{n_\alpha}{h} F_\alpha = 2 \sum_{\beta, j} h^{j/2-1} \int \rho^q \langle B_\alpha, c_{\alpha\beta j} (Z_j B_\beta + B_\beta Z_j g) \rangle.$$

The normalized nonhorizontal blocks have zero boundary value at $h = 0$ for smooth data: the moment forcing has been removed, and each remaining source has at least one spatial derivative and an additional positive power $h^{j/2}$ after time integration. Thus $F_\alpha(0) = 0$.

Lemma 3. *For every $\alpha = (i, k) \neq (1, 1)$ there is a polynomial $K_\alpha(a_1, \dots, a_{k-1})$ with nonnegative coefficients such that*

$$(5.4) \quad F_\alpha(1) + \int_0^1 [\mathcal{D}_q(B_\alpha) + h^{-1} F_\alpha] dh \leq K_\alpha(a_{<k}) E^2 \left[Q(t, 0) + \sum_{j < k} J_j \right].$$

The constants are independent of $a_k, \dots, a_s$, of $q \in (1, 3/2]$, and of all coefficient derivatives.

Proof. For a nonhorizontal input block β , integrate its source in (5.3) by parts:

$$(5.5) \quad \int \rho^q \langle B_\alpha, c(Z_j B_\beta + B_\beta Z_j g) \rangle = - \int \rho^q \langle Z_j B_\alpha + \varepsilon B_\alpha Z_j g, c B_\beta \rangle.$$

Using (2.6) and Young's inequality, this is bounded, after including $2h^{j/2-1}$, by a small part of $\mathcal{D}_q(B_\alpha)$ plus

$$C \|c\|^2 a_j^{-1} h^{-1} F_\beta.$$

For the horizontal input block, instead use $|B_0| \leq \sqrt{r}E$ and the positive damping $n_\alpha F_\alpha / h$. Its source is bounded by a small part of that damping plus

$$C \|c\|^2 h^{j-1} \int \rho^q (|Z_j B_0|^2 + rE^2 |Z_j g|^2).$$

Upon integration, (5.1) controls the first term and the second is $C \|c\|^2 rE^2 J_j$. All j occurring here are below k . Absorb the chosen small parts in (5.3), integrate, and induct in the strictly increasing total degree. Since $a_j \geq 1$, inverse a_j factors may be bounded by one. Squaring the finitely many source coefficients and multiplying by previously obtained constants produces a polynomial only in $a_{<k}$. This proves (5.4). $\square$

Define the physical remainder

$$(5.6) \quad \mathcal{E}_h = N - \mathbf{M}_h((\rho B_0)_H), \quad E_{ik} = h^{-(i+k-2)/2}(\mathcal{E}_h)_{ik}/\rho.$$

By the polynomial scaling of $\mathbf{M}_h$, each E_{ik} is a finite linear combination of lower normalized blocks. Thus, enlarging the polynomials,

$$(5.7) \quad \int_0^1 \frac{1}{h} \int \rho^q |E_{ik}|^2 dh \leq K_k(a_{<k}) E^2 \left[Q(t, 0) + \sum_{j < k} J_j \right].$$

Again there is no dependence on current or higher weights in K_k .

6. ENGEL: THE CORRECTION IN EXPLICIT COORDINATES

For $[X, Y] = Z$, $[X, Z] = W$, $[Y, Z] = 0$, with W central, use

$$\mathcal{A}_h = \frac{1}{2}(X^2 + Y^2) + \frac{1}{2}a_2 h Z^2 + \frac{1}{2}a_3 h^2 W^2.$$

Write the full tensor as

$$N = \begin{pmatrix} H_{11} & H_{12} & u_1 & v_1 \\ H_{12} & H_{22} & u_2 & v_2 \\ u_1 & u_2 & z & r \\ v_1 & v_2 & r & w \end{pmatrix}.$$

Put $V_1 = (\mathcal{C}_X^2 + \mathcal{C}_Y^2)/2$ and $V_2 = \mathcal{C}_Z^2/2$. All products of total raising degree at least six vanish. Hence

$$\mathbf{M}_h = I + hV_1 + \frac{1}{2}h^2(V_1^2 + a_2 V_2), \quad \mathbf{M}_h^{-1} = I - hV_1 + \frac{1}{2}h^2(V_1^2 - a_2 V_2).$$

The renormalized blocks are

$$(6.1) \quad \begin{aligned} \hat{H} &= H, & \hat{u} &= u, \\ \hat{v} &= v - \frac{h}{2}(H_{12}, H_{22})^T, & \hat{z} &= z - h \operatorname{tr} H, \\ \hat{r} &= r - \frac{3}{2}hu_2, & \hat{w} &= w - h(v_2 + z) + \frac{1}{4}h^2(2H_{11} + 3H_{22}) - \frac{1}{2}a_2 h^2 H_{11}. \end{aligned}$$

Normalize $\hat{u}, \hat{v}, \hat{z}, \hat{r}, \hat{w}$ by, respectively, $\sqrt{h}\rho, h\rho, h\rho, h^{3/2}\rho, h^2\rho$. The layer-two energy has a constant independent of both a_2, a_3 ; the layer-three energy has a constant depending on a_2 but independent of a_3 . This is exactly the strict dependency required for choosing the weights.

For horizontal constant input A , the positive deterministic moment tensor is

$$\mathbf{M}_h(A_H) = \begin{pmatrix} A_{11} & A_{12} & 0 & hA_{12}/2 \\ A_{12} & A_{22} & 0 & hA_{22}/2 \\ 0 & 0 & h \operatorname{tr} A & 0 \\ hA_{12}/2 & hA_{22}/2 & 0 & h^2(2a_2 A_{11} + 2A_{11} + 3A_{22})/4 \end{pmatrix}.$$

Thus all renormalized nonhorizontal blocks vanish for spatially constant m, A . The nonintegrable geometric variance in an uncorrected running-scale energy has been removed, not estimated in absolute value.

7. PHYSICAL-TIME COUPLING AND CLOSURE

Put $\mathcal{U}(t) = \int_0^1 Q(t, h) dh$. Split the exact physical flux as

$$\rho_t = \mathcal{H}(\mathbf{M}_h((\rho B_0)_H)) + \mathcal{H}\mathcal{E}_h.$$

We now estimate both parts. All parameter choices in this section precede the choice of ε .

7.1. The positive moment part. In the scaled covector frame $\widehat{g}_j = h^{(j-1)/2} \nabla_j g$, put $M = M_1((B_0)_H)$. Positivity of M_1 gives $M \geq 0$, $M_{11} = B_0 \geq I$, and

$$M_{kk} \leq EP_k(a_{<k})I$$

for fixed polynomials P_k . Positivity and a block Cauchy–Schwarz estimate give

$$(7.1) \quad \langle M\widehat{g}, \widehat{g} \rangle \geq \frac{1}{2} |\nabla_H g|^2 - C_{\mathbb{G}} E \sum_{k \geq 2} P_k(a_{<k}) h^{k-1} |\nabla_k g|^2.$$

After one integration by parts, the other term differentiates B_0 only. The norm of the fixed linear map M_1 , (5.1), and $J_* \leq 2J_1$ give

$$(7.2) \quad \left| \int_0^1 \int \rho^q \sum_{\alpha, \beta} \widehat{g}_\alpha [M_1((\widehat{Z}_\beta B_0)_H)]_{\alpha\beta} \right| \leq \frac{1}{16} J_1 + C_a Q(t, 0),$$

where $\widehat{Z}_\beta = h^{(d_\beta-1)/2} Z_\beta$ and C_a is finite and independent of q and coefficient derivatives. Dependence of C_a on all weights is harmless: it multiplies $q\varepsilon Q(t, 0)$, and ε is chosen last. Lemma 2 makes the higher-layer sum in (7.1) at most $J_1/16$.

7.2. The corrected tensor remainder. This is where two integrations by parts, rather than a pointwise tensor bound, are decisive. For a block with degrees $i \leq k$, its contribution to Q_t is a finite sum of terms

$$(7.3) \quad q\varepsilon \int \rho^q h^{(i+k-2)/2} E_{ik}(Z_k(Z_i g) + \varepsilon(Z_k g)(Z_i g)),$$

plus terms with $[Z_i, Z_k]g$. Indeed, interchange the two derivatives in the symmetrized Hessian so that the higher-layer derivative is outside; the resulting bracket has degree $i+k$.

Let $\mathfrak{E}_{ik} = \int_0^1 h^{-1} \int \rho^q |E_{ik}|^2 dh$. Equation (2.6) and Cauchy–Schwarz in space and scale give

$$(7.4) \quad \left| \int_0^1 \text{the } (i, k) \text{ remainder in } Q_t dh \right| \leq C_{\mathbb{G}} q\varepsilon \sqrt{\mathfrak{E}_{ik}} \left(\sqrt{L_i/a_k} + \sqrt{J_{i+k}} \right),$$

where $J_{i+k} = 0$ if $i+k > s$. The weights are exact: $h^{(i+k-2)/2} = h^{-1/2} h^{(i+k-1)/2}$.

Using (5.7) and the scalar lemma, the first part of (7.4) is bounded by

$$C_{\mathbb{G}} q\varepsilon E \sqrt{K_k(a_{<k})/a_k} \sqrt{Q(t, 0) + J_1} \sqrt{J_1}.$$

Make $E^2 K_k(a_{<k})/a_k$ small. For the bracket terms prescribe in Lemma 2 the finitely many polynomials

$$\widetilde{P}_l(a_{<l}) = E^2 \sum_{i+k=l} K_k(a_{<k}).$$

Their dependency is strictly lower because $k < i+k=l$. Choose their weighted sum times J_l small. The lemma and its simultaneous ratio conditions give one finite choice of weights satisfying all these requirements. Consequently all remainder terms together are bounded by

$$(7.5) \quad q\varepsilon \left(\frac{1}{16} J_1 + C_a Q(t, 0) \right),$$

after enlarging C_a .

7.3. The closed entropy inequality. Combining (7.1)–(7.5) gives

$$(7.6) \quad \mathcal{U}'(t) \leq -\frac{1}{4} q\varepsilon J_1 + C_a q\varepsilon Q(t, 0).$$

All constants and smoothing weights have already been fixed. From (2.9) and $J_* \leq 2J_1$,

$$Q(t, 0) - Q(t, 1) \leq q\varepsilon C_{\text{gap}} J_1, \quad C_{\text{gap}} = \max_j a_j.$$

Choose

$$(7.7) \quad 0 < \varepsilon \leq \min \left\{ \frac{1}{2}, \frac{1}{D}, \frac{1}{12C_{\text{gap}}(1+C_a)} \right\}.$$

Then

$$(7.8) \quad \boxed{\mathcal{U}'(t) + \frac{1}{8C_{\text{gap}}}Q(t, 0) \leq \frac{1}{4C_{\text{gap}}}Q(t, 1).}$$

The coefficient error is absorbed only now, by choosing $q - 1$ small. No smoothing weight was chosen using a coefficient derivative or $1/(q - 1)$.

8. ADJOINT ESTIMATE AND THE ELLIPTIC MAXIMUM ESTIMATE

By mass preservation and (2.2),

$$Q(t, h) \leq C_{\mathbb{G}}^{\varepsilon} h^{-D\varepsilon/2}.$$

Integrate (7.8) on $0 < t < 1$, use $\mathcal{U}(1) \geq 0$, and note that $D\varepsilon/2 < 1$. It follows that

$$(8.1) \quad \|P_t^* m_0\|_{L^q((0,1) \times \mathbb{G})} \leq C(\mathbb{G}, E, q) \|m_0\|_1.$$

This is first proved for smooth A and good positive initial data, uniformly in all coefficient derivatives, and then for nonnegative L^1 initial data by approximation and adjoint L^1 contraction. The same proof holds for every smaller $\varepsilon > 0$ than the fixed choice in (7.7).

8.1. Analytic justification for the smooth calculation. For smooth A equal to I outside a compact set, the operator L_A is a smooth horizontal diffusion satisfying Hörmander's condition. Its conservative positive semigroup and smooth solutions are standard smooth-coefficient inputs; no quantitative constant from such results is used in (7.8). For example, smooth coefficients are included in the fundamental-solution theory in [4, 5].

Conservativity can also be checked directly: in exponential coordinates the coefficients of the equation for a coordinate in layer k are polynomials in strictly lower-layer coordinates, multiplied by bounded entries of A or its square root. The first-layer process has bounded covariance. Induction in k and stopped moment estimates give finite moments on every finite time interval and rule out explosion. This argument does not bound a derivative of A .

For the a priori calculation one may take a smooth positive initial density with sufficiently fast polynomial decay and with its derivatives bounded by an integrable polynomial weight. Smooth-coefficient kernel bounds and local regularity, or the Duhamel formula against the fixed sub-Laplacian outside the compact perturbation, justify spatial cutoffs and all integrations by parts. The scalar and finite triangular tensor scale equations have the same admissible decay. At $h = 0$, the renormalized higher blocks tend to zero at a positive power of h for smooth inputs. Coefficient derivatives are used only qualitatively for these endpoint limits. None enters the bounds proved before passing to the limits. Initial densities in this class approximate arbitrary nonnegative L^1 data. These whole-group arguments avoid any lattice issue.

8.2. Resolvent duality and comparison. For smooth nonnegative compactly supported a , duality in (8.1) gives

$$\left\| \int_0^1 P_t a \, dt \right\|_{\infty} \leq C \|a\|_{p_0}, \quad p_0 = 1 + 1/\varepsilon.$$

Positivity and L^{∞} contraction give, by splitting the time integral into unit intervals,

$$(8.2) \quad v = (1 - L_A)^{-1} a = \int_0^{\infty} e^{-t} P_t a \, dt \geq 0, \quad \|v\|_{\infty} \leq C_R \|a\|_{p_0}, \quad L_A v = v - a.$$

The solution is smooth by smooth hypoellipticity. Let x_H denote first-layer coordinates. On B_1 choose

$$b_0 = (K - |x_H|^2)/(2r) > 0.$$

Then $L_A b_0 = -\text{tr } A/r \leq -1$. The usual interior-maximum argument applied to $w - \tau b_0$ proves comparison on every open subset of B_1 , without boundary regularity: at an interior maximum the full Euclidean gradient vanishes and $L_A(w - \tau b_0) \leq 0$.

8.3. Measurable coefficients and the active set. After subtracting the upper boundary value, fix $\delta > 0$ and $O_\delta = \{u > \delta\} \Subset \Omega$. Choose $\chi \in C_c^\infty(\Omega)$, $0 \leq \chi \leq 1$, equal to one near $\overline{O_\delta}$ and supported in $\{u > \delta/2\}$. Extend and mollify A to obtain smooth A_ν equal to I outside a fixed compact set, with $I \leq A_\nu \leq EI$ and local convergence in every finite L^p . Set $a_\nu = \chi(-L_{A_\nu}u)^+$. Then

$$(8.3) \quad \|a_\nu\|_{p_0} \leq \|f^{-1}\mathbf{1}_{\{u > \delta/2\}}\|_{p_0} + o(1).$$

The error on the fixed compact support is bounded by $|A_\nu - A| |D_H^{2,*}u|$; this tends to zero in L^{p_0} .

Approximate each continuous compactly supported a_ν uniformly from above by smooth non-negative functions supported in a fixed slightly larger compact set. Let $v = (1 - L_{A_\nu})^{-1}a$ and $M = \|v\|_\infty$. On O_δ ,

$$L_{A_\nu}(u - \delta - v - Mb_0) \geq -a - (v - a) + M \geq 0,$$

and the boundary values are nonpositive. Comparison and (8.2) give

$$\sup_{O_\delta}(u - \delta) \leq (1 + \|b_0\|_\infty)C_R\|a\|_{p_0}.$$

Pass through the smooth majorant approximation, then $\nu \rightarrow \infty$, then $\delta \downarrow 0$. This proves the normalized maximum estimate with the active-set indicator. It uses no existence or uniqueness assertion for a rough-coefficient resolvent or adjoint evolution.

Finally use $\Phi(g) = g_0\delta_R g$, $\bar{A} = \lambda^{-1}A \circ \Phi$, $\bar{u} = u \circ \Phi$. The forcing rescales as

$$\|\lambda^{-1}R^2f \circ \Phi\|_p = \lambda^{-1}R^{2-D/p}\|f\|_p.$$

For $p \geq p_0$, use Hölder's inequality and $|\Omega| \leq C_{\mathbb{G}}R^D$. This proves Theorem 1.

9. SUMMARY OF THE NONCIRCULAR DEPENDENCIES

At tensor layer k , (5.4) and (5.7) depend only on $a_1, \dots, a_{k-1}$. The current weight a_k supplies the denominator in (7.4); higher weights remain available for the bracket term of degree $i + k$. The scalar lemma handles any finite list of these strictly lower-dependent polynomials. The coefficients of $Q(t, 0)$ may depend on all weights, because $q - 1$ is chosen only after the weights have been fixed. This separates geometric absorption from coefficient absorption and closes the scale entropy.

The argument uses neither an assumed rough-coefficient Gaussian kernel nor a pointwise full-rank bound on the transported tensor. The replacement for the kinetic Gaussian is the scalar all-layer kernel (2.2), its exact tensor lift (4.2), and the finite moment correction (4.5).

REFERENCES

- [1] *Kinetic ABP through adjoint estimate*, supplied unpublished manuscript, `kinetic ABP-adjoint.tex`.
- [2] *A short analytic proof of a Heisenberg ABP estimate*, supplied unpublished manuscript, `heisenberg-abp.tex`.
- [3] N. Garofalo and G. Tralli, *A universal heat semigroup characterisation of Sobolev and BV spaces in Carnot groups*, arXiv:2205.04574, 2022.
- [4] A. Bonfiglioli, E. Lanconelli and F. Uguzzoni, *Fundamental solutions for non-divergence form operators on stratified groups*, Trans. Amer. Math. Soc. **356** (2004), 2709–2737.
- [5] M. Faini, *Non-divergence evolution operators modeled on Hörmander vector fields with Dini continuous coefficients*, Mathematics in Engineering **8** (2026), 477–517, doi:10.3934/mine.2026015.